

Data Analysis with R and Python

Midterm Solutions

Question: Let $Z = X_1 + X_2 + X_3$, where X_1, X_2, X_3 are independent exponential random variables with mean 1. Let μ denote the population median of Z . Compute the exact value of μ , rounded to four digits after the decimal point.

The distribution of Z is Gamma with shape parameter 3 and scale parameter 1. The corresponding median can be calculated using `qgamma()`:

```
qgamma(0.5, shape = 3, scale = 1)
```

```
[1] 2.67406
```

As a sanity check, we can confirm this using simulation:

```
replicate(10000, sum(rexp(3))) |> median()
```

```
[1] 2.685391
```

Question: A statistician who does not know the answer to question 1 above wants to estimate μ using simulation. He proposes the following procedure: Simulate i.i.d. observations $Z_1, Z_2, \dots, Z_{100}$, and then report the interval $(Z_{(40)}, Z_{(60)})$ as an interval estimate of μ . Here $Z_{(i)}$ denotes the i th smallest observation out of $Z_1, Z_2, \dots, Z_{100}$.

The statistician claims that although he does not know μ exactly, the procedure he has used guarantees a very high probability that μ falls inside the interval $(Z_{(40)}, Z_{(60)})$.

Estimate this probability p by replicating the procedure described above 10000 times, and reporting the number of replications in which $\mu \in (Z_{(40)}, Z_{(60)})$.

It is simplest to first write a function to simulate this experiment once.

```
sim_once <- function(mu) {  
  z <- rexp(100) + rexp(100) + rexp(100)  
  z <- sort(z)  
  z[[40]] < mu && mu < z[[60]]  
}
```

We can then count the number of times this experiment returns TRUE out of 10000 replications.

```
true_median <- qgamma(0.5, shape = 3, scale = 1)  
replicate(10000, sim_once(true_median)) |> sum()
```

```
[1] 9570
```

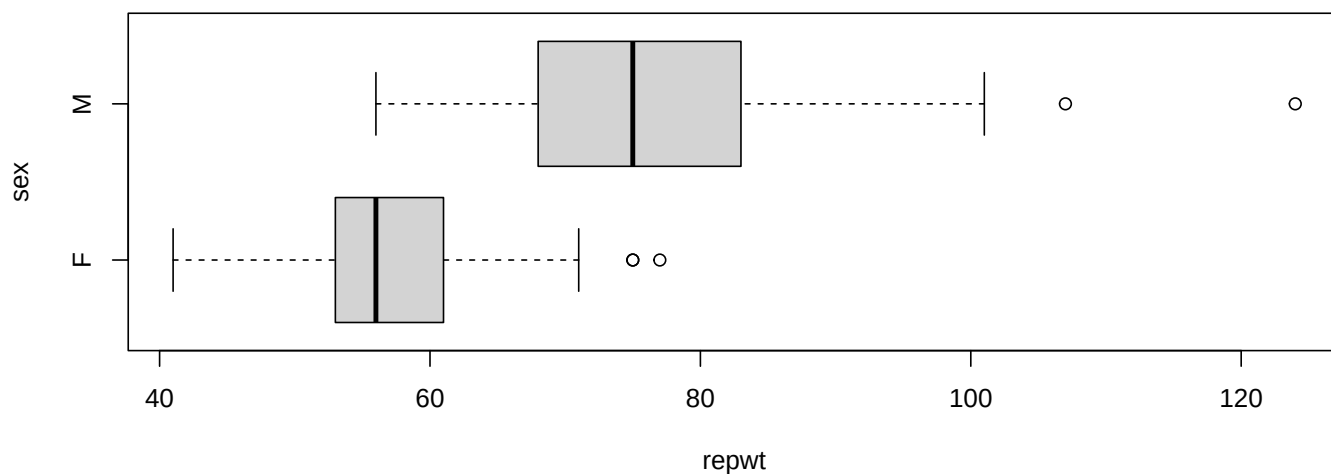
It can be shown that the exact probability is given by $P(40 \leq X < 60)$ where $X \sim \text{Bin}(100, \frac{1}{2})$. This exact probability is

```
pbinom(59, 100, 0.5) - pbinom(39, 100, 0.5)
```

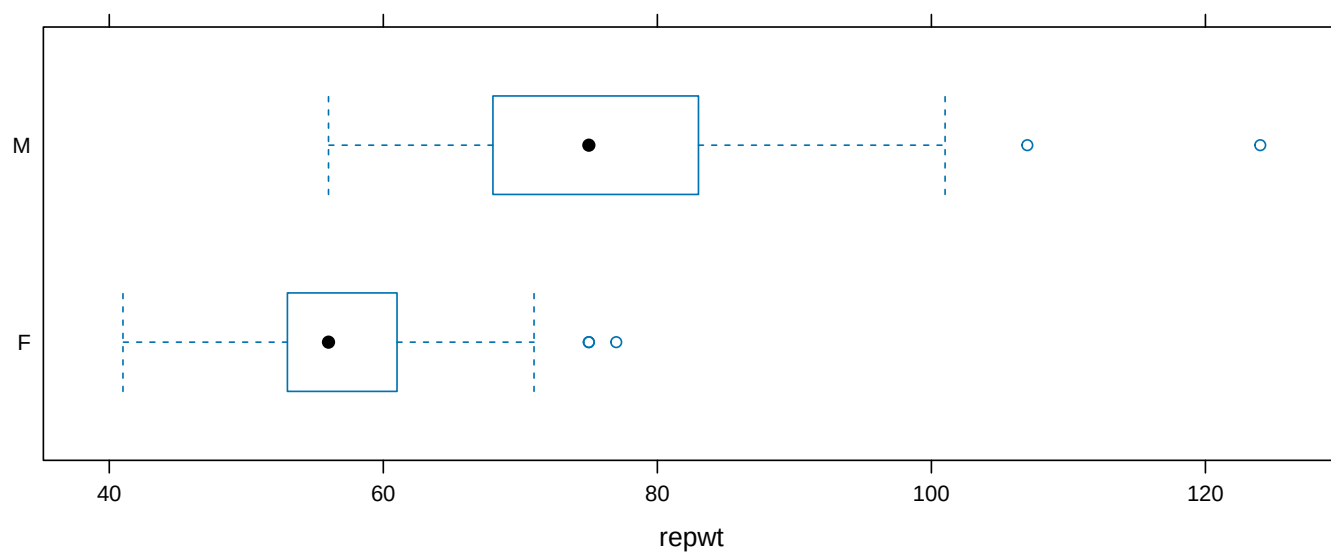
```
[1] 0.9539559
```

Question: Using the *Davis* dataset in the *carData* R package, draw comparative box-and-whisker plots of the variable *repwt* (self-reported weight) by *sex*.

```
library(carData)
graphics::boxplot(repwt ~ sex, data = Davis, horizontal = TRUE)
```



```
lattice::bwplot(sex ~ repwt, data = Davis)
```



Consider the following observations $X_1, X_2, \dots, X_{15}$ obtained in a survey:

140, 61, 73, 50, 158, 67, 64, 28, 386, 201, 17, 86, 99, 60, 165

Question: What is the value of θ which minimizes the loss function $\lambda(\theta) = \sum_{i=1}^n (X_i - \theta)^2$?

The value of θ which minimizes the sum of squared errors is the sample mean, which in this case is 110.333

Question: What is the value of θ which minimizes the loss function $\lambda(\theta) = \sum_{i=1}^n |X_i - \theta|$?

The value of θ which minimizes the sum of absolute errors is the sample median, which in this case is 73.

Question: An analyst wants to model the conditional distribution of infant mortality (y) on percentage of urban population (x), based on data for 193 countries collected around 2010. A scatter plot of the data is shown on a separate page.

Based on the plot, how should the analyst model the relationship of y on x ? If you are unsure, what are the next steps you would suggest to the analyst.

This is of course an open-ended question. However, one implicit assumption in simple linear regression is that the conditional variance of the response y given the predictor x does *not* depend on x . This is clearly not true in the plot of the original data.

The simplest solution in such cases is to try log or simple power transformations of the data. This does not always work, but it helps quite often, and is therefore the first thing one should try.

For this data, the following transformation seems to make the conditional variances much more reasonable. It also seems to make the relationship quite linear, which is an added bonus.

```
library(carData)
plot(log(infantMortality) ~ pctUrban, UN,
     xlab = "Per cent of population that is urban",
     ylab = "log(Infant deaths by age 1 year per 1000 live births)")
```

